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### Wind turbine control architecture

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#### Abstract

A wind turbine control architectures, including a turbine control portion which is configured to control at least one component of a wind turbine, and a turbine data storage portion which is configured to store a data storage container therein is provided. The turbine control portion and the turbine data storage portion communicate with each other via an inter-process communication.

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## Background/Summary

### CROSS-REFERENCE TO RELATED APPLICATIONS

(1) This application claims priority to PCT Application No. PCT/EP2020/076434, having a filing date of Sep. 22, 2020, which claims priority to EP Application No. 19203525.1, having a filing date of Oct. 16, 2019, the entire contents both of which are hereby incorporated by reference.

### FIELD OF TECHNOLOGY

(2) The following relates to a wind turbine control architecture and to a wind turbine comprising such a wind turbine control architecture.

### BACKGROUND

(3) FIG. 3 shows a wind turbine control architecture **101** according to the conventional art. The conventional art turbine control architecture **101** consists of two computers, that is a main computer **110** which is responsible for turbine control operations, and an interface computer **120**.

(4) The main computer **110** is responsible for collecting data from a wind park using I/O-stations **115**, which measure the turbine environment, and controls the necessary electric and mechanic devices for the optimized harvesting of the wind energy. The interface computer **120** uses a protocol converter for a communication between the main computer **110** and remaining systems in the wind park such as a SCADA system. The interface computer **120** also implements a SQL data storage container which stores some of the field values collected by the main computer **110** and makes these logged values available for an offline analysis and troubleshooting.

(5) The interface computer **120** is present in each wind turbine and used to store field data collected by the FO-stations **115** by the main computer **110**. The main computer **110** and the interface computer **120** communicate by an ethernet network **113**, where the main computer **110** samples the data and sends these values over the ethernet network **115** to the interface computer **120** which stores the data into the SQL data storage container.

(6) Due to the communication via the ethernet network **115** between both computers **110**, **120** and a legacy proprietary protocol being used, there is a jitter in the data from when they are sampled until they are logged-in by the interface computer **120**. Furthermore, the data can be lost due to an instability in the ethernet network **115**.

(7) Both computers **110**, **120** inside the wind turbine need to be setup and maintained. Besides the maintenance and initial costs, the wind turbine configuration is complex since the turbine control architecture **101** needs to be configured on the ethernet network **115** and the right set of permissions needs to be setup in both computers **110**, **120**.

### SUMMARY

(8) An aspect relates to a wind turbine control architecture which enables a more reliable turbine control under reduced costs.

(9) According to a first aspect of embodiments of the invention, a wind turbine control architecture comprises a turbine control portion which is configured to control at least one component of a wind

turbine, and a turbine data storage portion which is configured to store a data storage container therein. The data storage container is not necessarily a physical container and can comprise a database. The turbine control portion and the turbine data storage portion communicate with each other via an inter-process communication.

(10) The turbine control portion can be responsible for the data acquisition which collects data from the field using external devices such as I/O-stations and provide the values for turbine control operations and to the turbine data storage portion. In the wind turbine control architecture, the data is internally transferred using the inter-process communication so that the external conventional art ethernet network **115** (FIG. 3) or an ethernet interface for the data transfer can be omitted.

Furthermore, the data storage functionality is moved to the turbine data storage portion so that the conventional art interface computer can be eliminated. The costs are thus reduced in the investment stage and during the lifetime of the turbine.

(11) In addition, jitter and latency on the stored data can be reduced since the physical machine storing the data is the same which is acquiring the data from the field. With an improved data resolution, the diagnostics can be further improved and real-time analytics from the field data can be realized. The possibility of data loss due to network issues is removed. Higher sampling rates further allow better data resolution and thus better analytics, and preventive maintenances can be avoided.

(12) In an embodiment, the turbine control portion and the turbine data storage portion are integrally formed in a single wind turbine control device, for example a single computer. With the integration of the turbine control portion and the turbine data storage portion in a single computer, the conventional art interface computer **120** (FIG. 3) can be omitted by moving its functionality into the turbine control portion of embodiments of the present invention. In this wind turbine control architecture, the data storage container can be realized by the turbine data storage portion which uses a time-series data storage container design for data storage. As a result, the initial investment, and the cost of maintenance on the wind turbine lifetime are reduced.

(13) Furthermore, by having both the turbine control portion and the turbine data storage portion in the same computer, data loss can be avoided due to network problems or cabling errors. Since the turbine control portion and the turbine data storage portion are in the same computer, bandwidth issues, that are usually a bottleneck in the network, can also be removed since the data does not need to leave the computer.

(14) In an embodiment, the turbine data storage portion is configured to store a time-series data storage container, TSDB. By adding the time-series data storage container to the wind turbine control architecture, it is possible to store operational data in the same computer that is sampling the field data, whereby eliminating latencies in the data samples. The time-series data storage container further offers advantages in a relatively small data space and improved real-time analytics and diagnostics as well as troubleshooting as any event represented by a group of parameters is allocated to a predetermined time.

(15) In an embodiment, the turbine control portion is configured to communicate with at least one external device, for example the I/O-stations in the field.

(16) In an embodiment, the inter-process communication is a local process communication. In an embodiment, the inter-process communication uses a User Datagram Protocol (UDP).

(17) In an embodiment, the turbine control portion and the turbine data storage portion use a shared memory. A shared memory can be a memory that may be simultaneously accessed by multiple programs with an intent to provide communication among them or avoid redundant copies. Shared memory is an efficient passing data between programs. Depending on context, programs may run on a single processor or on multiple separate processors. Using memory for communication inside a single program, e.g., among its multiple threads, is also referred to as shared memory. A plurality of processes can share a certain part of a background memory such as a RAM (Random-Access Memory). For all processes involved, this shared memory area can be located in their address

spaces and can be read and changed with normal memory access operations. Since processors of the turbine control portion and the turbine data storage portion can share a single view of data and the communication between their processors is faster as memory accesses to the same location, the shared memory is easy to program. As an alternative to the shared memory concept, a distributed memory and a distributed shared memory concept can also be used.

(18) In an embodiment, the turbine control portion and the turbine data storage portion are connected by a local or internal bus which does not use the ethernet protocol.

(19) In an embodiment, the turbine control portion and the turbine data storage portion are configured to be setup and maintained together. The costs are thus reduced during the setup and during the lifetime of the wind turbine.

(20) In an embodiment, the turbine control portion and the turbine data storage portion are configured to directly interface data of the data storage container, and the turbine control portion is configured to impose rules and guarantees of Quality of Service (QoS) parameters.

(21) According to a second aspect of embodiments of the invention, a wind turbine is provided which comprises the wind turbine control architecture according to any one of the preceding claims.

(22) It must be noted that embodiments of the invention have been described with reference to different subject matters. Some embodiments have been described with reference to apparatus type claims whereas other embodiments have been described with reference to method type claims. However, a person skilled in the art will gather from the above and the following description that, unless other notified, in addition to any combination of features belonging to one type of subject matter also any combination between features relating to different subject matters, between features of the apparatus type claims and features of the method type claims is considered as to be disclosed with this application.

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## Description

### BRIEF DESCRIPTION

(1) Some of the embodiments will be described in detail, with reference to the following figures, wherein like designations denote like members, wherein:

(2) FIG. 1 shows an example of a wind turbine and different elements thereof;

(3) FIG. 2 shows a wind turbine control architecture according to an embodiment of the present invention; and

(4) FIG. 3 shows a wind turbine control architecture according to the conventional art.

### DETAILED DESCRIPTION

(5) FIG. 1 shows an example of a wind turbine **1**. The wind turbine **1** comprises a nacelle **3** and a tower **2**. The nacelle **3** is mounted at the top of the tower **2**. The nacelle **3** is mounted rotatable with regard to the tower **2** by a yaw bearing. The axis of rotation of the nacelle **3** with regard to the tower **2** is referred to as the yaw axis.

(6) The wind turbine **1** also comprises a hub **4** with three rotor blades **6** (of which two rotor blades **6** are depicted in FIG. 1). The hub **4** is mounted rotatable with regard to the nacelle **3** by a main bearing **7**. The hub **4** is mounted rotatable about a rotor axis of rotation **8**.

(7) The wind turbine **1** furthermore comprises a generator **5**. The generator **5** in turn comprises a rotor **10** connecting the generator **5** with the hub **4**. The hub **4** is connected directly to the generator **5**, thus the wind turbine **1** is referred to as a gearless, direct-driven wind turbine. Such a generator **5** is referred to as direct drive generator **5**. As an alternative, the hub **4** may also be connected to the generator **5** via a gear box. This type of wind turbine **1** is referred to as a geared wind turbine. Embodiments of the present invention are suitable for both types of wind turbines **1**.

(8) The generator **5** is accommodated within the nacelle **3**. The generator **5** is arranged and

prepared for converting the rotational energy from the hub **4** into electrical energy in the shape of an AC power.

(9) FIG. **2** shows a wind turbine control architecture **100** according to an embodiment of the present invention. The wind turbine control architecture **100** comprises a turbine control portion **10** which is configured to control at least one component of a wind turbine **1**. For example, the turbine control portion **10** can be configured to control a pitch angle of the rotor blades **6** and/or an output power of the generator **5**; however, embodiments of the present invention are not limited thereto.

(10) The wind turbine control architecture **100** further comprises a turbine data storage portion **11** which is configured to store a data storage container **12** therein. The turbine control portion **10** and the turbine data storage portion **11** communicate with each other via an inter-process communication **13**.

(11) The turbine control portion **10** can be responsible for the data acquisition which collects data from the field using external devices such as I/O-stations **15** and provide the values for turbine control operations and to the turbine data storage portion **11**. In the wind turbine control architecture **100**, the data is internally transferred using the inter-process communication **13** so that the external conventional art ethernet network **113** (FIG. **3**) for the data transfer can be omitted. Furthermore, the data storage functionality is moved to the turbine control portion **10** so that the conventional art interface computer **120** (FIG. **3**) can be eliminated. The costs are thus reduced in the investment stage and during the lifetime of the wind turbine **1**.

(12) In addition, jitter and latency on the stored data can be reduced since the physical machine storing the data is the same which is acquiring the data from the field. With an improved data resolution, the diagnostics can be further improved and real-time analytics from the field data can be realized. The possibility of data loss due to network issues is removed. Higher sampling rates further allow better data resolution and thus better analytics, and preventive maintenances can be avoided.

(13) The turbine control portion **10** and the turbine data storage portion **11** are integrally formed in a single wind turbine control device **14**. Such a wind turbine control device **14** can be embodied as a computer. With the integration of the turbine control portion **10** and the turbine data storage portion **11** in a single computer, the conventional art interface computer **120** (FIG. **3**) can be omitted by moving its functionality into the turbine control portion **10** of embodiments of the present invention. In this wind turbine control architecture **100**, the data storage container **12** can be realized by the turbine data storage portion **11** using a time-series data storage container design for data storage. As a result, the initial investment costs and the costs of maintenance during the lifetime of the wind turbine **1** are reduced.

(14) Furthermore, by having both the turbine control portion **10** and the turbine data storage portion **11** in the same computer **14**, data loss can be avoided due to network problems or cabling errors. Since the turbine control portion **10** and the turbine data storage portion **11** are in the same computer **14**, bandwidth issues, that are usually a bottleneck in a network, can also be removed since the data does not need to leave the computer **14**.

(15) The turbine data storage portion **11** is configured to store a time-series data storage container, TSDB **12**. By adding the time-series data storage container **12** to the wind turbine control architecture **100**, it is possible to store operational data in the same computer **14**, that is sampling the field data, whereby eliminating latencies in the data samples. The time-series data storage container **12** further offers advantages in a relatively small data space and improved real-time diagnostics and troubleshooting as any event represented by a group of parameters is allocated to a predetermined time.

(16) The time-series data storage container **12** can be implemented by a SQL database.

(17) The turbine control portion **10** is configured to communicate with at least one external device **15**, for example the I/O-stations **15** in the field.

(18) The inter-process communication **13** is a local process communication. In an embodiment, the

inter-process communication **13** can use a User Datagram Protocol (commonly abbreviated as UDP).

(19) The turbine control portion **10** and the turbine data storage portion **11** can use a shared memory. A shared memory can be a memory that may be simultaneously accessed by multiple programs with an intent to provide communication among them or avoid redundant copies. Shared memory is an efficient passing data between programs. Depending on context, programs may run on a single processor or on multiple separate processors.

(20) Using memory for communication inside a single program, e.g., among its multiple threads, is also referred to as shared memory. A plurality of processes can share a certain part of a background memory such as a RAM (Random-Access Memory). For all processes involved, this shared memory area can be located in their address spaces and can be read and changed with normal memory access operations. Since processors of the turbine control portion **10** and the turbine data storage portion **11** share a single view of data and the communication between their processors is faster as memory accesses to the same location, the shared memory is easy to program.

(21) As an alternative to the shared memory concept, a distributed memory and a distributed shared memory concept can also be used.

(22) The turbine control portion **10** and the turbine data storage portion **11** are configured to be setup and maintained together. The costs are thus reduced during the setup and during the lifetime of the wind turbine **1**.

(23) The turbine control portion **10** and the turbine data storage portion **11** are connected by a local or internal bus (not shown). Such an internal bus can comprise an internal data bus, a memory bus, a system bus or a front-side-Bus. The system bus can comprise an address bus, a control bus and a data bus.

(24) The turbine control portion **10** and the turbine data storage portion **11** are configured to directly interface data of the data storage container **12**, and the turbine control portion **10** is configured to impose rules and guarantees of Quality-of-Service (QoS) parameters such as (bit) rate, reliability, packet loss, throughput, transmission delay, availability, jitter and security of data flow.

(25) Although the present invention has been disclosed in the form of preferred embodiments and variations thereon, it will be understood that numerous additional modifications and variations could be made thereto without departing from the scope of the invention.

(26) For the sake of clarity, it is to be understood that the use of “a” or “an” throughout this application does not exclude a plurality, and “comprising” does not exclude other steps or elements.

## Claims

1. A wind turbine control architecture for a wind park, comprising: a single wind turbine control device provided with a turbine control device and a turbine data storage device therein; wherein the turbine control device is configured to control at least one component of a wind turbine, wherein the turbine control device collects data from the wind park; and wherein the turbine data storage device is configured to store a time-series data storage container, TSDB, therein, wherein the time-series data storage container, TSDB, is implemented by a SQL database; wherein the turbine control device and the turbine data storage device communicate with each other via an inter-process communication, wherein the turbine control device and the turbine data storage device are integrally formed and structurally incorporated within the single wind turbine control device.
2. The wind turbine control architecture according to claim 1, wherein the turbine control device is configured to communicate with at least one external device.
3. The wind turbine control architecture according to claim 1, wherein the inter-process communication is a local process communication.
4. The wind turbine control architecture according to claim 1, wherein the inter-process

communication uses a User Datagram Protocol.

5. The wind turbine control architecture according to claim 1, wherein the turbine control device and the turbine data storage device use a shared memory, and wherein the turbine control device and the turbine data storage device share a portion of background memory.
  6. The wind turbine control architecture according to claim 1, wherein the turbine control device and the turbine data storage device are connected by a local or internal bus.
  7. The wind turbine control architecture according to claim 1, wherein the turbine control device and the turbine data storage device are configured to be setup and maintained together.
  8. The wind turbine control architecture according to claim 1, wherein the turbine control device and the turbine data storage device are configured to directly interface data of the time-series data storage container, TSDB; and the turbine control device is configured to impose rules and guarantees of Quality of Service, QoS, parameters.
  9. A wind turbine comprising the wind turbine control architecture according to claim 1.
  10. The wind turbine control architecture according to claim 1, wherein the turbine control device is configured to control at least two components of a wind turbine.
  11. The wind turbine control architecture according to claim 1, wherein the time-series data storage container, TSDB, comprises a physical container.
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